

Hyperbolic trig identities

Basic skills

Q1.

$$a) y = \cosh x = \frac{e^x + e^{-x}}{2} \quad b) y = \operatorname{cosech} x = \frac{2}{e^x - e^{-x}}$$

$$c) y = \sinh x = \frac{e^x - e^{-x}}{2} \quad d) y = \operatorname{sech} x = \frac{2}{e^x + e^{-x}}$$

$$e) y = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad f) y = \coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

Q2. $\tanh x = \frac{\sinh x}{\cosh x}$ is a definition

$$\begin{aligned} \text{Q3. } \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{e^{2x} + 2 + e^{-2x} - (e^{2x} - 2 + e^{-2x})}{4} = \frac{4}{4} = 1 \end{aligned}$$

Q4. $\sinh^2 x - 2\sinh x - 3 = 0$

$$\Rightarrow (\sinh x - 3)(\sinh x + 1) = 0$$

$$\text{So } \sinh x = 3$$

or

$$\sinh x = -1$$

$$\Rightarrow x = \ln(3 + \sqrt{3^2 + 1}) \quad x = \ln(-1 + \sqrt{1^2 + 1})$$

$$x = \ln(3 + \sqrt{10})$$

$$\text{arc sinh } x = \ln(x + \sqrt{x^2 + 1})$$

$$x = \ln(\sqrt{2} - 1)$$

Competency skills

$$\text{Q5. } \operatorname{sech}^2 x + \tanh^2 x = \left(\frac{2}{e^x + e^{-x}} \right)^2 + \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)^2$$

$$= \frac{4 + e^{2x} - 2 + e^{-2x}}{(e^x + e^{-x})^2} = \frac{e^{2x} + 2 + e^{-2x}}{(e^x + e^{-x})^2}$$

$$= \frac{(e^x + e^{-x})^2}{(e^x + e^{-x})^2} = 1$$

$$\text{Q6. } \cosh^2 x - \sinh^2 x = 1$$

$$\Rightarrow \frac{\cosh^2 x}{\cosh^2 x} - \frac{\sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x}$$

$$\Rightarrow 1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\Rightarrow \operatorname{sech}^2 x + \tanh^2 x = 1$$

Q7. a) $\cosh 2x - 8 \sinh x = 3$

$\downarrow \cosh 2x = \cosh^2 x + \sinh^2 x$

$\Rightarrow \cosh^2 x + \sinh^2 x - 8 \sinh x = 3$

$\downarrow \cosh^2 x = 1 + \sinh^2 x$

$1 + 2\sinh^2 x - 8 \sinh x - 3 = 0$

$\Rightarrow \sinh^2 x - 4 \sinh x - 1 = 0$

$\Rightarrow \sinh x = \frac{4 \pm \sqrt{16 + 4}}{2} = 2 \pm \sqrt{5}$

so $x = \ln \left(2 \pm \sqrt{5} + \sqrt{(2 \pm \sqrt{5})^2 + 1} \right)$
 $= \ln \left(2 \pm \sqrt{5} + \sqrt{8 \pm 4\sqrt{5}} \right)$

b) $\sinh^2 x = 5 \cosh x + 2$

so $\cosh^2 x - 1 = 5 \cosh x + 2$

$\Rightarrow \cosh^2 x - 5 \cosh x - 3 = 0$

$\Rightarrow \cosh x = \frac{5 \pm \sqrt{25 + 12}}{2} = \frac{5 \pm 2\sqrt{10}}{2}$

now $5 - 2\sqrt{10} < 0$ so only

$\cosh x = \frac{5 + 2\sqrt{10}}{2} = 2.5 + \sqrt{10}$

$$\text{So } x = \ln (2.5 + \sqrt{10} \pm \sqrt{(2.5 + \sqrt{10})^2 - 1})$$

$$= \ln (2.5 + \sqrt{10} \pm \sqrt{15.25 + 5\sqrt{10}})$$

c) $3 \operatorname{sech}^2 x = \tanh x$

$\checkmark \operatorname{sech}^2 x + \tanh^2 x = 1$

$$3(1 - \tanh^2 x) = \tanh x$$

$$\Rightarrow 3 \tanh^2 x + \tanh x - 3 = 0$$

$$\text{So } \tanh x = \frac{-1 \pm \sqrt{1 + 36}}{6} = \frac{-1 \pm \sqrt{37}}{6}$$

$$\text{As } \frac{-1 - \sqrt{37}}{6} < -1 \text{ only } \tanh x = \frac{-1 + \sqrt{37}}{6}$$

$$\text{So } x = \frac{1}{2} \ln \left(\frac{1 + \frac{-1 + \sqrt{37}}{6}}{1 - \left(\frac{-1 + \sqrt{37}}{6} \right)} \right)$$

$$= \frac{1}{2} \ln \left(\frac{+ \sqrt{37}}{12 - \sqrt{37}} \right)$$

d) $\coth^2 x - \frac{2}{\sinh x} = 2$

$$\Rightarrow \coth^2 x - 2 \operatorname{cosech} x = 2$$

$\coth^2 x - \operatorname{cosech}^2 x = 1$

$$\text{SO } 1 + \operatorname{cosech}^2 x - 2 \operatorname{cosech} x - 2 = 0$$

$$\operatorname{cosech}^2 x - 2 \operatorname{cosech} x - 1 = 0$$

$$(\operatorname{cosech} x - 1)^2 = 0$$

$$\Rightarrow \operatorname{cosech} x = 1 \Rightarrow \sinh x = 1$$

$$\Rightarrow x = \ln(1 + \sqrt{1^2 + 1}) = \ln(1 + \sqrt{2})$$

Q8.

$$\tanh 2x = \frac{\sinh 2x}{\cosh 2x}$$

$$= \frac{2 \sinh x \cosh x}{\cosh^2 x + \sinh^2 x} = \frac{2 \frac{\sinh x}{\cosh x}}{1 + \frac{\sinh^2 x}{\cosh^2 x}}$$

→ $\div \cosh^2 x$ on top + bottom

$$= \frac{2 \tanh x}{1 + \tanh^2 x}$$

$$\text{Q9. } \cosh 3x = \frac{e^{3x} + e^{-3x}}{2}$$

$$= \frac{(e^x + e^{-x})^3 - 3e^x - 3e^{-x}}{2}$$

$$= 4 \left(\frac{e^x + e^{-x}}{2} \right)^3 - 3 \left(\frac{e^x + e^{-x}}{2} \right)$$

$$= 4 \cosh^3 x - 3 \cosh x$$

Advanced skills

$$\text{Q10. } \sinh 3x = \frac{e^{3x} - e^{-3x}}{2}$$

$$= \frac{(e^x - e^{-x})^3 - 3e^x + 3e^{-x}}{2}$$

$$= 4 \left(\frac{e^x - e^{-x}}{2} \right)^3 - 3 \left(\frac{e^x - e^{-x}}{2} \right)$$

$$= 4 \sinh^3 x - 3 \sinh x$$

$$\text{Q11. } \sinh(A+B) = \frac{e^{A+B} - e^{-(A+B)}}{2}$$

$$= \frac{e^A e^B - e^{-A} e^{-B}}{2}$$

$$\begin{aligned}
 &= \frac{2e^A e^B + 2\bar{e}^{-A}(e^B - \bar{e}^B) + 2\bar{e}^{-B}(e^A - \bar{e}^A) - 2\bar{e}^{-A}\bar{e}^{-B}}{2 \times 2} \\
 &= \frac{(e^A - \bar{e}^{-A})(e^B + \bar{e}^B) + (e^A + \bar{e}^{-A})(e^B - \bar{e}^B)}{4} \\
 &= \frac{e^A - \bar{e}^{-A}}{2} \frac{e^B + \bar{e}^B}{2} + \frac{e^A + \bar{e}^{-A}}{2} \frac{e^B - \bar{e}^B}{2} \\
 &= \sinh A \cosh B + \cosh A \sinh B
 \end{aligned}$$

Q12. $a \sinh x - b \cosh x = c$

a) If $c \neq 0$ only identity linking $\sinh$, $\cosh$ is $\tanh x$ but would give $\operatorname{sech} x$ as well

b) $a \left(\frac{e^x - \bar{e}^{-x}}{2} \right) - b \left(\frac{e^x + \bar{e}^{-x}}{2} \right) = c$

$$(a-b)e^x - (a+b)\bar{e}^{-x} = 2c$$

$$(a-b)e^{2x} - 2ce^x - (a+b) = 0$$

$$e^x = \frac{2c \pm \sqrt{4c^2 + 4(a-b)(a+b)}}{2(a-b)}$$

$$e^x = \frac{c \pm \sqrt{c^2 + a^2 - b^2}}{a - b}$$

$$\text{so } x = \ln \left(\frac{c \pm \sqrt{c^2 + a^2 - b^2}}{a - b} \right)$$

$$c) R \sinh(x + \alpha) = R \sinh x \cosh \alpha + R \cosh x \sinh \alpha$$

$$\Rightarrow R \cosh \alpha = a, \quad R \sinh \alpha = -b$$

$$\Rightarrow R^2 \cosh^2 \alpha - R^2 \sinh^2 \alpha = a^2 - b^2$$

$$\Rightarrow R^2 = a^2 - b^2 \Rightarrow R = \pm \sqrt{a^2 - b^2}$$

$$\text{now } \tanh \alpha = -\frac{b}{a} \quad \alpha = \frac{1}{2} \ln \left(\frac{1 - \frac{b}{a}}{1 + \frac{b}{a}} \right)$$

$$\text{so } \sqrt{a^2 - b^2} \sinh \left(x + \frac{1}{2} \ln \left(\frac{1 - \frac{b}{a}}{1 + \frac{b}{a}} \right) \right) = c$$

$$\Rightarrow \sinh \left(x + \frac{1}{2} \ln \left(\frac{a - b}{a + b} \right) \right) = \frac{c}{\sqrt{a^2 - b^2}}$$

$$x + \frac{1}{2} \ln \left(\frac{a - b}{a + b} \right) = \ln \left(\frac{c}{\sqrt{a^2 - b^2}} + \sqrt{\frac{c^2}{a^2 - b^2} + 1} \right)$$

$$2C = \ln \left(\frac{C + \sqrt{C^2 + a^2 - b^2}}{\sqrt{a^2 - b^2}} \right) - \ln \left(\frac{\sqrt{a-b}}{\sqrt{a+b}} \right)$$

$$= \ln \left(\frac{\frac{C + \sqrt{C^2 + a^2 - b^2}}{\sqrt{a^2 - b^2}}}{\frac{\sqrt{a-b}}{\sqrt{a+b}}}} \right)$$

4 $\rightarrow \sqrt{a-b} \sqrt{a+b}$

$$= \ln \left(\frac{C + \sqrt{C^2 + a^2 - b^2}}{a-b} \right)$$